1 **(a)** $068 + 180$ **[M1]**

248 **[A1]**

(b) 270 **[B1]**

[Total Marks: 3]

2 **(a)** $128 + 180$ **[M1]**

308 **[A1]**

(b) 52 **[B1]**

(c) $180 - 90 - 52$ **[M1]**

38 **[A1]**

[Total Marks: 5]

3 **(a)** $180 + 148$ OR $360 - (180 - 148)$ **[M1]**

328 **[A1]**

(b) **(i)** 270 **[B1]**

(ii) Identifies the bearing of P from B is $090°$ **[B1]**

$148 - 90 = 58$ **[A1]**

[Total Marks: 5]

4 **(a)** Bearing of P from $Q = 115 + 180 = 295°$ **[M1]**

Angle $PQR = 295° - 205° = 90°$ **[A1]**

(b) $15^2 + 8^2$ **[M1]**

$225 + 64 = 289$ **[M1]**

17 **[A1]**

[Total Marks: 5]

5 **(a)** 150 **[B1]**

(b) $180 + 65$ OR $360 - (180 - 65)$ **[M1]**

245 **[A1]**

(c) Identifies bearing of A from B is $245°$ (can follow through from (b)) **[M1]**

$245 - 70 = 175$ **[A1]**

(d) 25 **[B1]**

[Total Marks: 6]

6 **(a)** Interior angle at B (South to BA) $= 35°$ OR interior angle from North line $= 180 - 35 = 145°$ **[M1]**

Angle $ABC = 360 - 145 - 125 = 90°$ OR $125 - 35 = 90°$ **[A1]**

(b) $AC^2 = 12^2 + 16^2$ **[M1]**

$AC^2 = 144 + 256 = 400$ **[M1]**

$AC = 20$ **[A1]**

(c) $\tan(\angle BAC) = \dfrac{16}{12}$ or $\sin(\angle BAC) = \dfrac{16}{20}$ **[M1]**

$\angle BAC = 53.13°$ **[A1]**

Bearing $= 035° + 53.13° = 088.1°$ **[A1]**

[Total Marks: 8]

7 **(a)** $\angle QPR = 118° - 42° = 76°$ **[M1]**

$QR^2 = 24^2 + 35^2 - 2(24)(35)\cos(76°)$ **[M1]**

$QR = 37.3$ or $37.34\ldots$ **[A1]**

(b) $\dfrac{\sin(\angle PQR)}{35} = \dfrac{\sin(76°)}{37.34}$ OR $\cos(\angle PQR) = \dfrac{24^2 + 37.34° - 35^2}{2(24)(37.34)}$ **[M1]**

$\sin(\angle PQR) = 0.909\ldots$ OR $\cos(\angle PQR) = 0.415\ldots$ **[M1]**

$\angle PQR = 65.4°$ or $65.42\ldots°$ **[A1]**

Bearing $= 180 + 42 - 65.4 = 156.6°$ (or $157°$) **[A1]**

[Total Marks: 7]

8 **(a)** Bearing of P from $A = 050 + 180 = 230°$ **[M1]**

Angle $PAB = 230° - 160° = 70°$ **[M1]**

$PB^2 = 40^2 + 60^2 - 2(40)(60)\cos(70°)$ **[M1]**

59.6 or $59.65\ldots$ **[A1]**

(b) Uses sine rule or cosine rule to find angle APB **[M1]**

$\dfrac{\sin(APB)}{60} = \dfrac{\sin(70°)}{59.65}$ or $\cos(APB) = \dfrac{40^2 + 59.65^2 - 60^2}{2(40)(59.65)}$ **[M1]**

Angle $APB = 70.9°$ $(70.93\ldots°)$ **[A1]**

Bearing $= 050 + 70.9 = 120.9°$ **[A1]**

[Total Marks: 8]

9 **(a)** Identifies angle $PQR = 230° - 120° = 110°$ (or uses parallel lines to find $180 - 50 = 130$ and $360 - 130 - 120 = 110$) **[B1]**

Uses Cosine rule: $PR^2 = 45^2 + 60^2 - 2(45)(60)\cos(110°)$ [M2] (M1 for correct substitution into formula)

$PR = 86.4$ or $86.44\ldots$ **[A1]**

(b) Uses Sine rule: $\dfrac{\sin(QPR)}{60} = \dfrac{\sin(110°)}{86.44}$ **[M1]**

Angle $QPR = 40.7°$ $(40.73\ldots)$ **[A1]**

Bearing $= 050° + 40.7° = 090.7°$ **[A1]**

(c) $090.7 + 180$ **[M1]**

270.7 **[A1]**

[Total Marks: 9]

10 $048 + 180$ **[M1]**

228 **[A1]**

[Total Marks: 2]

11 (a) $124 + 180$ **[M1]**

304 **[A1]**

(b) $\angle PQR = 90°$ or $\angle QPR = 180° - 124° = 56°$ **[M1]**

$\angle PRQ = 180° - 90° - 56° = 34°$ **[A1]**

[Total Marks: 4]

12 (a) Bearing of A from B is $065° + 180° = 245°$ or Interior angle between North and AB is $180° - 65° = 115°$ **[M1]**

$245° - 155° = 90°$ or $360° - 115° - 155° = 90°$ **[A1]**

(b) Triangle ABC is right-angled and isosceles, so $\angle BCA = (180° - 90°) \div 2 = 45°$ **[M1]**

Bearing of B from C $= 155° + 180° = 335°$ **[M1]**

Bearing of A from C $= 335° - 45° = 290°$ **[A1]**

(c) $\angle DAB = 90°$ **[M1]**

Bearing of D from A $= 065° + 90° = 155°$ **[A1]**

[Total Marks: 7]